

A Distance Metric for Object-Centric Process Mining

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Abstract. Object-centric process mining offers a holistic view of real-life business processes by considering multiple interrelated objects simultaneously. Comparing these business objects is essential for uncovering patterns such as customer behavior or resource usage. To address this, analytical techniques like clustering and cohort identification are frequently applied, all of which rely on suitable distance measures. However, existing distance measures often lack critical properties to ensure a meaningful application to interacting business objects. As a result, distance-based analysis methods might lead to flawed analytical results in that object-centric setting. In this paper, we identify a set of five important requirements for distance measures tailored towards analysis techniques on interacting business objects. To satisfy all requirements, we introduce a novel distance measure that combines graph-based weights with existing Levenshtein distances in a cost-minimization framework. We benchmark our approach against existing methods, demonstrating its ability to uncover insights that alternative techniques cannot. Furthermore, we assess the runtime feasibility of our approach to ensure practical applicability.

Keywords: process mining · distance measures · object-centric

1 Introduction

In many organizations, users interact with information systems as part of business processes. These activities are often logged and aggregated into so-called *event logs*. These logs typically contain the executed activities themselves, the corresponding execution timestamps, and the executing resources. Process mining uses these logs to analyze business processes with the goal of generating business-relevant insights regarding the performance levels and the process costs.

Traditionally, process mining focuses on the life cycle of a single isolated business object, such as a purchase order. However, in practice, many real-life processes involve multiple interconnected business objects. For example, a purchase order might be associated with multiple items, resources, and various other related objects like the corresponding invoices, shipments, and customers.

Object-Centric Process Mining addresses this limitation by considering related business objects. Unlike traditional process mining, focusing on single-object life cycles, object-centric approaches recognize the complex interplay be-

tween business objects within a process. By analyzing these relationships, object-centric techniques provide a more comprehensive and accurate view.

A common analysis step for object-centric event logs is the identification of groups of objects that exhibit similar behavior. These groupings are often achieved through the application of clustering techniques and enable organizations to gain targeted insights. For example, sales teams can use such groupings to tailor their outreach strategies for different customer segments. This becomes especially important in large event logs, where high variability makes manual inspection impractical. Clustering techniques help break down the data into interpretable subgroups, revealing patterns and simplifying subsequent analysis.

However, these methods rely on a distance measure that quantifies behavioral differences between objects in a standardized and meaningful way. In the traditional process mining setting, distance measures only need to account for the behavior of isolated objects. However, for the successful application in the object-centric setting, object interactions need to be considered as well.

For real-life processes, two important factors play a crucial role in this regard, besides the isolated behavior of the individual objects themselves. First, the number of interactions an object has with other objects significantly impacts its importance. An object with many interactions plays a more central role, amplifying its potential effect on the overall process in case of potential performance issues. Second, the level of transitivity of the relationship between two objects is key. Directly interacting objects have a stronger and more immediate impact on each other compared to those that are only linked through intermediaries.

Existing distance measures largely disregard such properties. Some existing methods compute the distance between two objects by exclusively focusing on events related to those objects and their attribute values [1–3]. Other approaches do not consider the impact of, potentially transitively, interacting objects on each other at all [4,5]. As such, the generated results of various object-centric analysis methods that rely on such distance measure can be flawed. For example, when comparing orders, existing measures either ignore the properties of attached items or weight them equally to the order, so the order itself has little impact compared to its items.

In this paper, we identify five requirements for the application of distance measures on interacting business objects based on existing literature. To satisfy these requirements, we introduce a novel distance measure that combines graph-based weights with existing Levenshtein distances in a cost-minimization framework. We evaluate our approach by comparing clusters derived using our distance measure against clusters derived from existing distance measures. Additionally, we provide a feasibility analysis, where we analyze the runtime on multiple logs. While this paper highlights clustering as a key application, the proposed distance measure is broadly applicable and can support other analysis tasks such as anomaly detection, process monitoring, and optimization.

The remainder of this paper is organized as follows. Section 2 introduces the preliminaries. In Section 3, we outline the identified requirements for the application of an object-centric distance measure. Section 4 introduces our novel

measure based on graph-weights and Levenshtein distances that adheres to these requirements. Section 5 evaluates our approach in terms of feasibility and analytical value. Section 6 discusses related work. Section 7 concludes the paper.

2 Preliminaries and Background

Object-Centric Event Logs: An object-centric event log is an ordered sequence of events $L = \langle (a_1, O_1), (a_2, O_2), \dots \rangle$, where each a_i denotes an activity and each O_i is a set of involved objects [6]. We denote the sets of all activities and objects in a log by Σ and Θ respectively. While timestamps are not shown in this representation, they are assumed to determine the total order. That is, the sequence reflects the temporal order. Table 1 presents a running example log L_1 .

Object Graph Relations: Given an object-centric event log L , the object graph $G = (\Theta, E)$ is an undirected graph, representing the relations between objects based on their interactions [7]. The objects Θ form the vertices of the object graph. The edges are described by the set $E = \{(o_i, o_j) | (a, O) \in L \wedge o_i, o_j \in O\}$, which contains edges between two objects if they occur in the same object set. Figure 1 shows the object graph of the log L_1 . Two objects are called related if they transitively interact with each other in the graph. Formally, we write $o_1 \sim_{rel} o_2$ if there is a path from o_1 and o_2 within the object graph of L and $o_1, o_2 \in \Theta$. Additionally, We define $tra : \Theta \times \Theta \rightarrow \mathbb{N}$ as the shortest-path length between two objects in G , and $deg : \Theta \rightarrow \mathbb{N}$ as the node degree function

Ground Truth Distance Measure: A ground truth distance measure is a function $l : \Theta \times \Theta \rightarrow \mathbb{R}_{\geq 0}$ that quantifies the dissimilarity between the ordered activities of two objects o_1 and o_2 . It can be instantiated using any established control flow distance metric applied to the activity sequences associated with individual objects, as known from traditional (non-object-centric) event logs. In essence, l measures how similar the isolated behavior of the two objects is, based solely on their activities, independent of the broader object graph context.

Object Distance: An object distance measure is function $d_G : \Theta \times \Theta \rightarrow \mathbb{R}_{\geq 0}$ that quantifies the dissimilarity between two objects, potentially taking into account any available information in the object-centric event log.

Table 1. Object-centric event log with orders (O), items (I) and packages (P).

Create O - {O1, I1, I2},	Select I - {I1},	Modify I - {I1},
Select I - {I2},	Pack I - {I1, P1},	Pack I - {I2, P2},
Pay O - {O1},	Create O - {O2, I3, I4, I5},	Select I - {I3},
Select I - {I4},	Select I - {I5},	Pack I - {I3, I4, I5, P3},
Pay O - {O2}		

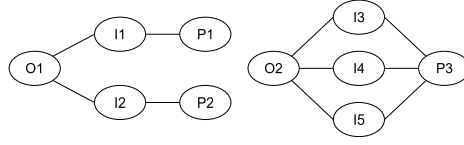


Fig. 1. Example object graph for L_1

Object Impact: We formalize the concept of an object impact within an object distance measure d_G . When measuring the distance between two objects, other objects might or might not impact the resulting value. We quantify that potential impact with a function $I_{d_G(o_1, o_2)} : \Theta \rightarrow \mathbb{R}$. This function quantifies the change in distance between o_1 and o_2 that is caused by removing $o \in \Theta$ from the log graph. If that impact is zero, it means that any change to o does not influence the distance between o_1 and o_2 . Formally, let there be an object graph $G = (O, E)$ with $o \in O$ and an equivalent object graph $G' = (O', E')$ with $o \notin O'$ where the impact of o is distributed proportionally to the objects in G' . If o has a higher ground distance l than the average, then $d_G(o_1, o_2) > d_{G'}(o_1, o_2)$. Conversely, if o has a smaller-than-average ground distance, then $d_G(o_1, o_2) < d_{G'}(o_1, o_2)$.

3 Distance Measure Requirements

In this section, we identify and formalize five important requirements for the application of distance measures on object-centric event logs. For the remainder of this paper, we treat those five requirements as axiomatic goals.

3.1 Requirement Identification

Our axiomatic requirements are grounded in established principles of business process analysis and a literature review. Specifically, we adopt the assumption that all objects related to a process entity are relevant, and that their influence can be modulated by their structural properties within the object graph.

First, we assume that all objects related to the compared entities should influence the distance measure. This reflects the notion of contextual completeness, as supported by Dumas et al. [8], who argue in their product-based process modeling approach that all connected information elements contribute to the derivation of a product or decision. Second, we assume that the impact of related objects should decay with increasing graph distance. In the same model, objects with shorter derivation paths are treated as more central to the execution logic, justifying a stronger influence on outcome. Third, we consider the number of connections an object has (its degree) as an indicator of its relevance. This is supported by Weske [9], who emphasizes that in data-driven process architectures, objects referenced by multiple activities, and therefore potentially more objects, are more critical to case execution. Both the product data model [8] and

the case-handling metamodel [9] reinforce the idea that proximity and connectivity determine the business relevance of process-related data objects.

Additionally, we performed a literature search for existing applications of distance-based analysis technique in the object-centric setting, and similar research domains with different names, such as artifact-centric process mining, to select requirements. In particular, we adopt the notion that any object-centric distance measure should provide the mathematical properties of a metric space.

3.2 Formal Axioms

Our first axiom ensures that every object related to one of the two compared objects influences the distance. This axiom ensures that the metric is sensitive to all contextually connected components. For L_1 , when comparing the two orders O_1 and O_2 , their associated items and packages must impact the measure.

Axiom A1. *Related objects should influence the distance measure: $\forall o \in O : o \sim_{rel} o_1 \vee o \sim_{rel} o_2 \implies I_{d_G(o_1, o_2)}(o) > 0$*

Additionally, we aim to ensure that with increasing transitivity, the impact of the related objects should decay. In our example log graph, the packages should have less of an impact on orders, than the directly interacting items.

Axiom A2. *The distance measure should be sensitive to the transitivity. That is, when calculating the distance $d_G(o_1, o_2)$, the related objects $\forall o_i, o_j \in O : o_i, o_j \sim_{rel} o_1$, with a higher number of edges $tra(o_i, o_1) > tra(o_j, o_1)$ and identical node degree $deg(o_i) = deg(o_j)$ have lower impact: $I_{d_G(o_1, o_2)}(o_i) < I_{d_G(o_1, o_2)}(o_j)$. An analogous condition holds with o_2 .*

Next, we ensure that objects with more connected objects should have a higher impact on the distance measure. As such, we require a higher node degree in the object graph to correspond to a higher impact of the corresponding object. Objects that are related to many other objects can exert significant influence across those connections. This axiom prevents a package associated with a single item from being handled in the same manner as one associated to many.

Axiom A3. *The distance measure should be sensitive to the degree. For $d_G(o_1, o_2)$ and $\forall o_i, o_j \in O : o_i, o_j \sim_{rel} o_1$ a higher degree $deg(o_i) > deg(o_j)$ with the same transitivity $tra(o_i, o_1) = tra(o_j, o_1)$ implies: $I_{d_G(o_1, o_2)}(o_i) > I_{d_G(o_1, o_2)}(o_j)$. An analogous condition holds with o_2 .*

Similar to traditional distance measures, we ensure that the isolated control flow of each object influences the distance measure. The individual control flow of an object defines its process behavior. Consequently, in the context of OCPM, comparing the control flow of objects requires also considering the control flows of their related objects, as these relationships can significantly influence the overall process dynamics from a business perspective.

Axiom A4. *The measure should be sensitive to the control flow. Let $G = (O, E)$ and $G' = (O', E')$ be two equivalent object graphs except for the control flow of $o_i \in O$ and $o'_i \in O'$. Then, $l(o_i, o_1) < l(o'_i, o_1)$ implies $d_G(o_1, o_2) < d_{G'}(o_1, o_2)$.*

Lastly, we require the distance measure to be a metric [10], which ensures properties such as non-negativity, the identity of indiscernibles, symmetry, and the triangle inequality. Interpreting the objects in a log as elements of a metric space allows us to apply tools from geometry and topology, such as clustering, multidimensional scaling, or nearest-neighbor search. For numerous analytical techniques—such as hierarchical clustering, k-means, and dimensionality reduction methods like t-SNE or UMAP—a true metric is a strict prerequisite, thereby broadening the scope and reliability of the measure. While future extensions may refine or expand this set, these five axioms capture the core properties.

Axiom A5. *The distance measure should be a metric.*

4 Distance Metric Construction

After identifying relevant requirements for object-centric distance measures, we introduce a distance metric that adheres to all of them. We do so, by first determining the impact of each object onto the distance measure for a given object pair. Then, we use the impacts to create a distance metric with the Levenshtein distance and the Earth-Movers Distance (EMD) [11].

4.1 Impact of related objects

For a given pair o_1, o_2 We define $I_{d_G(o_1, o_2)}(o) = 0$ for all objects o that are unrelated to both o_1 and o_2 . In all other cases, if $o \sim_{\text{rel}} o_1$, the influence is: $I_{d_G(o_1, o_2)}(o) = \frac{1}{2} (D_{\text{norm}}(o_1)[o] + C_{\text{norm}}(o_1)[o])$, where D_{norm} is a normalized measure based on graph distance, and C_{norm} is a normalized measure based on its connectivity. If $o \sim_{\text{rel}} o_2$, the same expression is used, substituting o_2 in place of o_1 . The detailed computation of D_{norm} and C_{norm} is provided in the following sections.

Impact of graph distance. We compute a list of object graph distances $D(o) = [(o_1, d_1) \dots (o_n, d_n)]$ by calculating $d_i = \text{tra}(o, o_i)$ for all objects $o_1 \dots o_n \sim_{\text{rel}} o$. As we want to have normalized values for the weights, we then compute $\text{norm}_1 = \sum_{i=1}^n \delta^{d_i}$ where n is the number of related objects and $\delta \in (0, 1)$ is a parameter which is introduced to scale the effect of the distance. It also ensures that the closer the objects are to o , the higher the normalized value gets. Finally, we get the list of normalized distance values for all related objects:

$$D_{\text{norm}}(o) = [(o_1, \frac{\delta^{d_1}}{\text{norm}_1}), (o_2, \frac{\delta^{d_2}}{\text{norm}_1}), \dots]$$

Impact of connectivity. We utilize the degree $\text{deg}(o)$ of an object o within the object graph G . As we again need to receive normalized values, we compute $\text{norm}_2 = \sum_{i=1}^n \text{deg}(o_i)$ where n is the number of related objects. Afterward, we

can compute $C_{norm}(o)$ to have a second impact value component for the node degree.

$$C_{norm}(o) = [(o_1, \frac{deg(o_1)}{norm_2}), (o_2, \frac{deg(o_2)}{norm_2}), \dots]$$

Example 1. For the object graph of $O1$, we have $D_{norm}(O1) = [(O1, 0.4), (I1, 0.2), (I2, 0.2), (P1, 0.1), (P2, 0.1)]$. Increasing δ amplifies the relative impact of objects that are further away from $O1$, i.e., with higher $tra(O1, o)$. Furthermore, we have $norm_2 = 8$ and therefore $C_{norm}(O1) = [(O1, 0.25), (I1, 0.25), (I2, 0.25), (P1, 0.125), (P2, 0.125)]$. For $P1 \sim_{rel} O1$, the impact $I_{d_G(O1, O2)}(P1)$ is 0.1125 for the calculation of the distance between $O1$ and $O2$.

4.2 Object Distance

Based on the impact factor for each related object, we construct a framework using the EMD to calculate a final distance value between two objects. The choice of the EMD relates to Axiom A5, which demands that the measure is a metric. For the EMD, we already know that it is a metric if the used ground truth distance is a metric as well [11]. One ground truth distance that can capture the process flow while still being a metric is the Levenshtein distance of two objects, for which we write $lv(o_i, o_j)$. While alternatives like Jaccard or DTW exist, they either ignore event order or violate metric properties. In the end, any metric ground distance is compatible with EMD—our approach does not depend on Levenshtein specifically. Additionally, the EMD is capable of finding a minimal distribution of two probability masses. The higher the mass of one node, the higher its impact on the overall distance is. As such, we can intuitively use $I_{d_G(o1, o2)}(o)$ as the probability mass for each related object.

We introduce the auxiliary variables $x_{i,j} \in [0, 1]$, which states how much of the impact of $o_i \sim_{rel} o_1$ is mapped to the impact of $o_j \sim_{rel} o_2$.

$$\begin{aligned} d_G(o_1, o_2) = \min_{x \geq 0} \quad & \sum_{i=1}^n \sum_{j=1}^m lv(o_i, o_j) x_{ij} \\ \text{s.t.} \quad & \sum_i x_{ij} = I_{d_G(o_1, o_2)}(o_j) \quad \forall j, \quad \sum_j x_{ij} = I_{d_G(o_1, o_2)}(o_i) \quad \forall i. \end{aligned}$$

The minimization problem illustrated presents an implementation of the EMD. The objective is to minimize the total transportation cost, where the cost between two objects is defined as the Levenshtein distance between the sequences of events associated with them. The two constraints ensure that the entire "impact" of each object is fully redistributed across the related objects. Since the total impact across all objects in both distributions (those related to o_1 and o_2 , respectively) sums up to one, a solution is guaranteed to be found [11].

Example 2. Figure 2 shows the flow of the impact to calculate the distance between the two objects $O1, O2$ from the running example. For example for $O1$: $I_{d_G(O1, O2)}(O1) = 0.325$. The distance between the orders given by this flow is $d_G(O1, O2) = 0.11$.

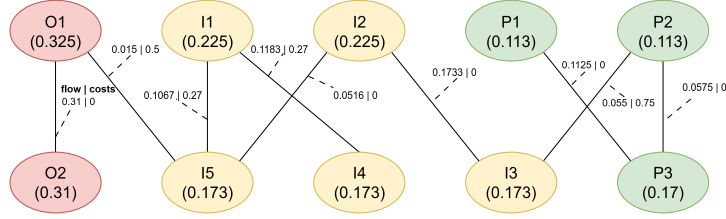


Fig. 2. Example EMD for $O1$, $O2$ from the running example

Theorem 1. *The constructed $d_G(o_1, o_2)$ satisfies the axioms A1 - A5*

We prove that the axioms are correctly implemented in our approach. For the correctness of the defined impact function $I_{d_G(o_1, o_2)}$, we can utilize the dual formulation of the proposed d_G to use a sensitivity analysis to demonstrate that reducing the impact of an object with above-average cost decreases the total matching cost. This implies that removing such an object reduces the overall distance, validating the design of the impact function. In particular, you can show that if an object o_t has a higher-than-average individual cost, then even a small reduction in its impact leads to a lower total cost due to the structure of the dual solution, confirming that its removal is beneficial and thus proving the correct implementation of the impact.

Moving on to the requirements of the distance measure itself, starting with Axiom A1, we know that every related object o is reachable from either o_1 or o_2 . As a result, there is at least one edge connected to every o . Therefore $D_{norm}(o_1)[o] > 0$ or $D_{norm}(o_2)[o] > 0$. The same argument holds for the function C_{norm} . As a result, $I_{d_G(o_1, o_2)}(o) > 0$ and the first axiom is guaranteed.

$I_{d_G(o_1, o_2)}(o)$ is the average of $D_{norm}(o)$ and $C_{norm}(o)$. If for two objects $o_i, o_j \sim_{rel} o_1$ they both have the same degree but $d_G(o_1, o_i) > d_G(o_1, o_j)$, then we know that $C_{norm}(o_1)[o_i] = C_{norm}(o_1)[o_j]$ and $D_{norm}(o_1)[o_i] < D_{norm}(o_1)[o_j]$, leading to $I_{d_G(o_1, o_2)}(o_i) < I_{d_G(o_1, o_2)}(o_j)$. As a result, Axiom A2 holds for our implementation of the impact. The proof of A2 proceeds analogously and is therefore omitted due to space constraints. Furthermore, if we modify the control flow of an $o_t \sim_{rel} o_1$, such that we increase all flow distances with at least ϵ , then the solution of our minimization problem also increases by at least $I_{d_G(o_1, o_2)}(o_t) \cdot \epsilon$. As a result, Axiom A4 holds for our introduced metric. Finally, we constructed the distance by using the EMD together with the Levenshtein distance as ground distance to ensure that A5 holds [11].

5 Evaluation

For the evaluation, we first perform a case study on a publicly available data set. We compare the results of a clustering algorithm using the proposed distance measure against currently available distance measures. Finally, we analyze the runtime of the distance computation on three logs.

Table 2. Clusters derived using d_G

	Cluster Description
0	Follows the P2P flow with one purchase, delivery, and payment per case.
1	Multiple goods receipts and invoices per PO, single payments.
2	Shows duplicate POs and goods receipts tied to a single requisition, indicating possible maverick buying or data issues.
3	No purchase requisitions approval
4	High number of goods receipts per PO, suggesting fragmented deliveries.
5	Distinct for using delegated purchase requisition approvals.
6	Multiple payment events for the same PO
7	Excessive invoice receipts and matches, pointing to processing loops.

5.1 Clustering in a Purchase-to-Pay Process

We use the distances computed by our metric to cluster purchase requisitions from the purchase-to-pay process described in [12]. According to the dataset description, the process includes unauthorized purchases that bypass the standard approval procedure, as well as duplicate payments, which lead to some invoices being paid multiple times. We aim to cluster the 927 purchase requisitions.

Using the clustering algorithm DBSCAN with $\epsilon = 0.05$ and our proposed distance, results in the 8 clusters described in Table 2. As there are two clusters (3, 5) that stand out for processes without purchase approval, we can find the behavior that there are unauthorized purchases. Additionally, as there is a unique cluster that has multiple payment events for the same PO (6) we find this insight detectable with our proposed metric as well. The maximal achievable silhouette score [13] for DBSCAN and this distance measure is 0.48.

Comparison. We compare our findings against two different approaches. Ghahfarokhi et al. create an object profile that contains information about its process flow, numerical attributes, and categorical attributes [3]. For our comparison, we focus on the process flow and omit the other attributes, leading to an approach that considers only the process flow of the unique object. Following this approach, we identify three clusters: **Cluster 0:** Create and approve purchase requisition. **Cluster 1:** Create, delegate, and approve purchase requisition. **Cluster 2:** Create request for quotation. In summary, we find that some purchase requisitions are not approved. However, this approach does not provide any insight into the duplicated execution of payments, as this behavior occurs across all clusters. As this approach is only based on the events of the purchase requisition, this approach has a silhouette score of 1.

Finally, we compare our findings with the approach of Faria Junior et al. [1]. In their method, a global object identifier is assumed to be available, which is then used with a trace distance metric such as Levenshtein. We use the purchase requisition as this global identifier. As a result, all objects have the same influence on the distance calculation, which leads to diverse and versatile clusters. These reflect a wide range of P2P process patterns. Some cases follow standard

Table 3. Feasibility of different datasets, for the related objects we provide the median.

Log	Compared type	# Pairwise Comp.	# Related obj.	Runtime
PTP	Purchase Requisition	429 201	8	130s
OTC [14]	Orders	1 999 0000	7	1 000s
Sustainable [15]	Customer	10 731	101	2 200s

flows with one purchase order, goods receipt, invoice, and payment—indicating minimal complexity. Others reveal more complex structures, such as partial deliveries, multiple-stage processing, or fragmented invoicing, often accompanied by consolidated payments. Certain process variants involve parallel handling of several purchase orders for a single requisition, sometimes tied to the same quotation, representing scenarios like partial fulfillment or split sourcing. Some clusters bypass the purchase requisition approval phase entirely, beginning the process at the quotation or purchase order stage, which suggests ad-hoc or urgent procurement. Still others are characterized by a high number of goods receipts and intricate invoice matching steps, indicating fragmented deliveries or billing corrections. While these clusters offer valuable insight into process variations, the specific behavior of duplicated payments appears in most clusters. As such, this particular anomaly cannot be reliably detected based on clustering results alone. The optimal silhouette score using DBSCAN of this distance is 0.39.

5.2 Feasibility

Table 3 gives the feasibility analysis for three selected OCEL logs. In addition to the above-presented purchase-to-pay log, we select an order-to-cash log with a higher number of comparable objects. Additionally, we select the sustainability log to also consider objects with a higher connectivity. Both parameters have an important impact on the runtime. However, performing millions of computations of the EMD is naturally time-consuming. To mitigate this, various heuristics and approximation methods can be employed to significantly reduce the computational load while still maintaining reasonable accuracy [16]. Performance of the metric, which depends on the number of connected components, can be enhanced by leveraging techniques like the advanced case notion to reduce the object graph [17].

6 Related Work

We have defined five axioms that are fulfilled by our proposed distance measure. In this section, we discuss related work and their shortcomings regarding these axioms for the application in the object-centric setting. In the approach of Ghahfarokhi et al., the object graph structure is no longer used, and therefore, this approach does not fulfill our requirements. This approach violates Axiom A1, as related objects will have no impact on the distance between objects [3].

Table 4. Related work with axioms

Paper	A1	A2	A3	A4	A5
Ghahfarokhi et al [3]	N	N	N	Y	Y
Faria Junior et al [1]	Y	N	N	Y	Y
Jalali et al [2]	N	N	N	N	N
Zager et al [4]	Y	N	N	N	N
Vitanyi et al [18]	Y	N	N	Y	Y
Trad. PM	N	N	N	-	-
Proposed distance	Y	Y	Y	Y	Y

The approach of Faria is not capable of differentiating between two related objects. Each related object has the same impact, violating Axioms A2,A3 [1].

Furthermore, in a paper by Jalali, two object types are compared by analyzing the likelihood of events based on the object type. By those insights, a distance measure for object types can be derived, but it is not possible to differentiate between different objects from one type [2]. In our approach, we aim to compare specific objects. Additionally, Axiom A5 is not satisfied.

The approach of Zager is not created for object-centric process mining. However, it can analyze the structure of an object graph by comparing the neighborhood of two nodes within two graphs [4]. While the object structure is considered this approach is not able to weigh related objects. Additionally, it does not consider the content of the nodes and therefore violates Axiom A4. Axiom A5 is violated because some non-isomorphic graphs can have a distance of zero.

The Normalized Information Distance is a universal measure based on Kolmogorov complexity, aiming to quantify the similarity between objects by assessing the length of the shortest possible program that can transform one object into another [18]. For our purpose, one could count the number of events and objects one has to modify or add to get to similar objects. This measurement would successfully implement Axiom A1, as every object would be considered in this transformation. However, it would make no difference how close a transformed object is. Therefore, Axiom A2 and Axiom A3 would not hold.

Additionally, there are several distance measures in traditional process mining. These distance measures are not all capable of implementing the axioms A1 - A3, as they do not consider the object structure in object-centric event logs. To the best of our knowledge, our approach is the first distance measure designed for the application of object-centric logs that adhere to the identified axioms.

7 Conclusion

In this paper, we identified five axioms for the application of distance measures in the specialized context of object-centric process mining. We proposed an approach to compute distances between objects that satisfies all of these axioms. In particular, we constructed a metric that considers the behavioral aspects of individual objects and the interactions between them in a balanced manner.

We applied our techniques to a publicly available object-centric event log. This evaluation showed that clustering algorithms using our proposed distance measure can identify several business-relevant clusters that can not be found by current distance metrics. These insights related to both, the behavioral perspective of individual objects as well as the interactions between them. Additionally, the runtime demands of the distance calculation remained feasible.

Future work will be focused on the identified limitations. We plan to explore the application of advanced case notions and heuristics for the EMD to improve runtime scalability in the case of input logs with large sets of interacting objects.

References

1. E. R. Faria Junior, T. R. Neubauer, M. Fantinato, and S. M. Peres. Clustering analysis and frequent pattern mining for process profile analysis: An exploratory study for object-centric event logs. In *ICPM Workshops*, LNBPI.
2. A. Jalali. Object type clustering using markov directly-follow multigraph in object-centric process mining. *IEEE Access*, 10:126569–126579, 2022.
3. A. F. Ghahfarokhi, F. A., F. Zandkarimi, and W. M. P. van der Aalst. Clustering object-centric event logs. In *DATA*, pages 444–451. SCITEPRESS, 2023.
4. L. A. Zager and G. C. Verghese. Graph similarity scoring and matching. *Applied Mathematics Letters*, 21(1):86–94, 2008.
5. H. Xu. An algorithm for comparing similarity between two trees. *CoRR*, abs/1508.03381, 2015.
6. A. Berti, G. Park, M. Rafiei, and W. M. P. van der Aalst. An event data extraction approach from SAP ERP for process mining. In *ICPM Workshops*, volume 433 of *LNBIP*, pages 255–267. Springer, 2021.
7. J. N. Adams, D. Schuster, S. Schmitz, G. Schuh, and W. M. P. van der Aalst. Defining cases and variants for object-centric event data. In *ICPM*. IEEE, 2022.
8. M. Dumas, M. La Rosa, J. Mendling, and Hajo A. Reijers. *Fundamentals of Business Process Management, Second Edition*. Springer, 2018.
9. M. Weske. *Business Process Management - Concepts, Languages, Architectures, 2nd Edition*. Springer, 2012.
10. M. Fréchet. Sur quelques points du calcul fonctionnel. 1906.
11. Z. Yu and G. Herman. On the earth mover’s distance as a histogram similarity metric for image retrieval. In *ICME*, pages 686–689. IEEE, 2005.
12. P. Gyunam and L.T. genannt Unterberg. Procure-to-payment (p2p) object-centric event log in ocel 2.0 standard, October 2023.
13. P. J. Rousseeuw. Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *J. Comput. Appl. Math.*, 20:53–65, 1987.
14. B. Knopp and W. M.P. van der Aalst. Order management object-centric event log in ocel 2.0 standard, September 2023.
15. M. Heinisch, N. Graves, and W. M. P. van der Aalst. socel 2.0: A sustainability-enriched ocel of a hinge production process, September 2024.
16. O. Pele and M. Werman. Fast and robust earth mover’s distances. In *2009 IEEE 12th ICCV*, pages 460–467. IEEE, 2009.
17. J. N. van Detten, P. Schumacher, and S. JJ Leemans. A framework for advanced case notions in object-centric process mining. In *ICPM*, pages 402–414, 2024.
18. P. M. B. Vitányi, F. J. Balbach, R. Cilibrasi, and M. Li. Normalized information distance. *CoRR*, abs/0809.2553, 2008.